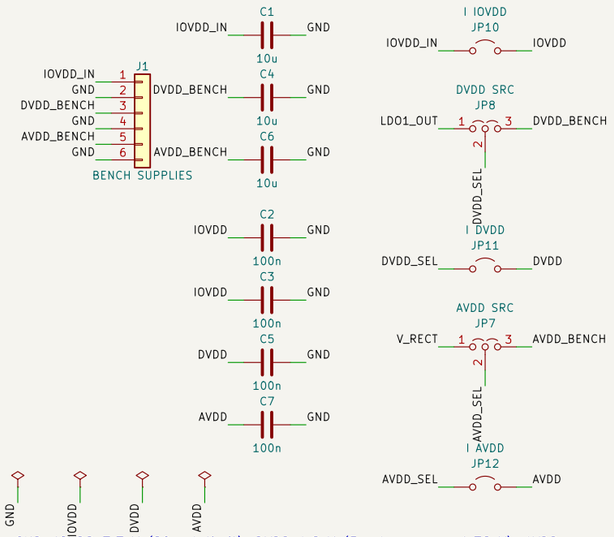
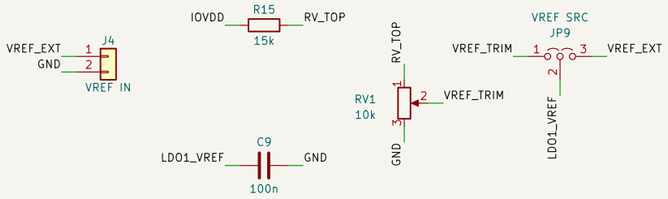
Supplies, source selection and current measurement



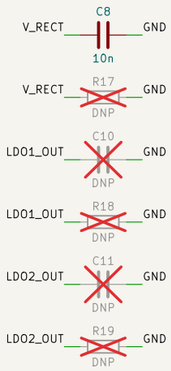
Power up GND, IOVDD 3.3 V (20 mA limit), DVDD 1.2 V (5 mA, never > 1.32 V), AVDD.
JP10–JP12: remove and insert a DMM to read each rail current.
JP7 / JP8: 1–2 = from the chip (V_RECT, LDO1_OUT), 2–3 = from the bench.

LDO_1 reference



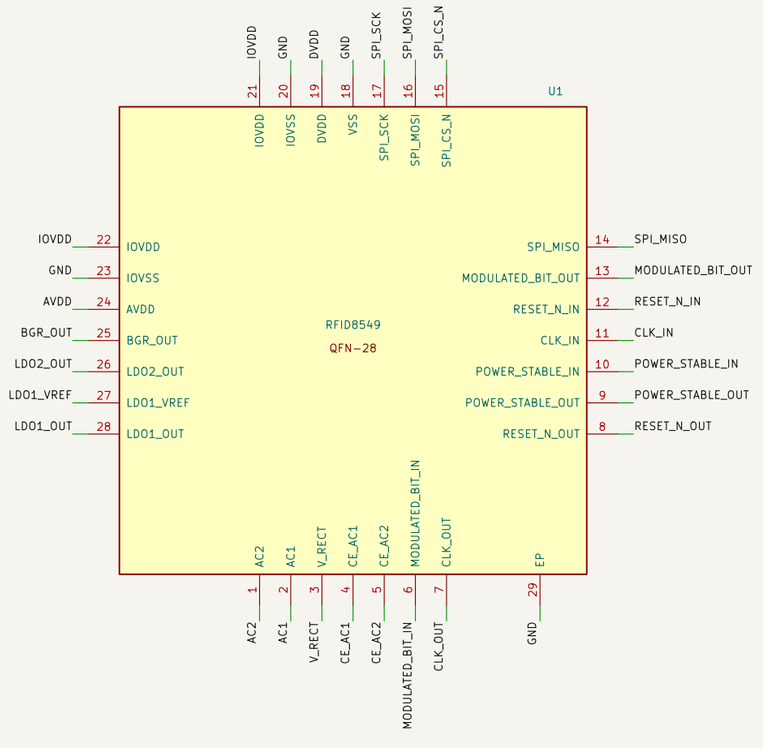
The bandgap output is not routed on the die: force LDO1_VREF = 1.000 V
from RV1 (JP9 1–2) or from J4 (JP9 2–3). LDO1_OUT = 1.2 x LDO1_VREF.

AFE output loads (values TBD with the analog team)



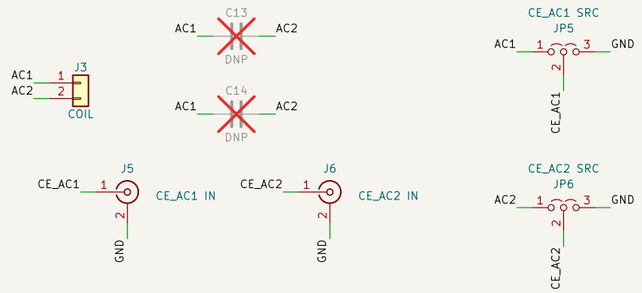
C8 = V_RECT storage capacitor (Cext). DNP parts are load footprints.

RFID8549 (QFN-28, 5 x 5 mm)



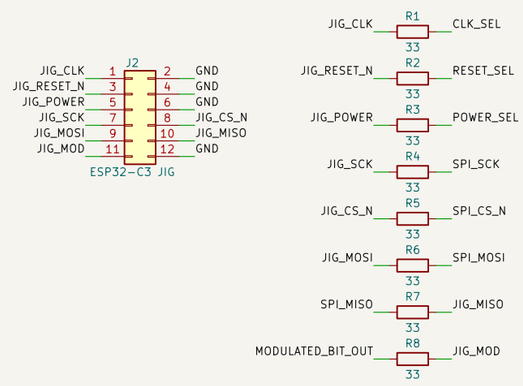
Pin 1 = S1 (AC2), bottom-left, counter-clockwise. EP to GND.
SPI_MISO tri-state pins are swapped on silicon: keep R9 pull-down, no pull-up anywhere.

Antenna and clock-extractor inputs



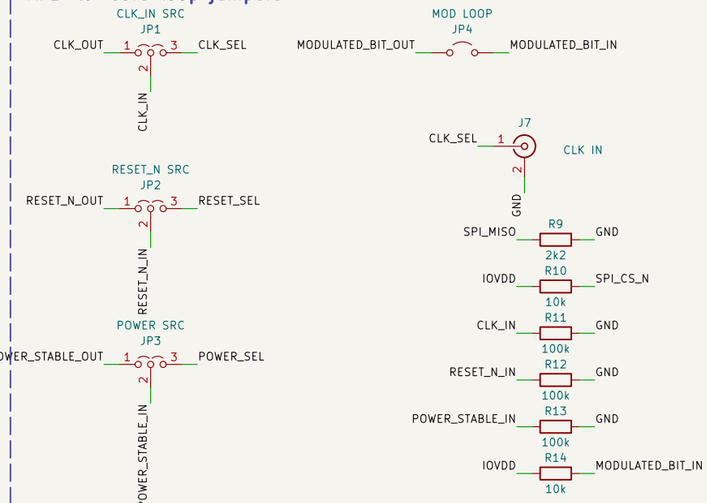
AC1/AC2 are bare pads, no ESD: drive J3 only from a floating source
(isolation transformer or a coil in a 134.2 kHz field). C13/C14: tuning, TBD.
JP5/JP6: 1–2 = AC to clock extractor (D6), 2–3 = park at GND. Remove to inject on J5/J6.

ESP32-C3 jig header (esp32c3/main/rfid_tag.h)



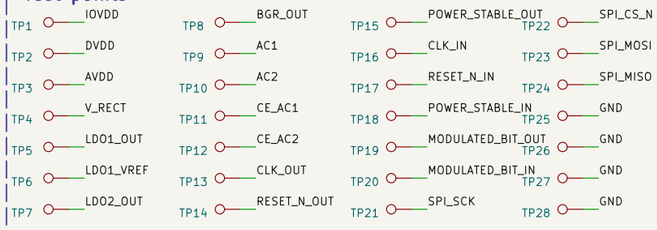
J2: 1 CLK GPIO0, 3 RESET_N GPIO1, 5 POWER GPIO3, 7 SCK GPIO4,
8 CS_N GPIO5, 9 MOSI GPIO6, 10 MISO GPIO7, 11 MOD GPIO10, 2/4/6/12 GND.

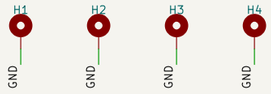
AFE-to-core loop jumpers



JP1–JP3: 1–2 = chip AFE drives the core (D6), 2–3 = jig drives it (D1–D5).
JP4 open = S6 held high, modulator off. J7: clock from a generator (unplug J2 pin 1).

Test points





Test plan: RFID8549 Digital Bring-up Test Scenario
Bring-up board for the RFID8549 LF RFID tag (IHP SG13G2), QFN-28
CDTA / University of Blida 1

Sheet: /
File: rfid8549_testboard.kicad_sch

Title: RFID8549 test board

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1